Física General II **()**

Solución

Instituto Tecnológico de Costa Rica

Selección única:

Pregunta	Respuesta
1	D
2	C
3	B
4	D

Problema 1 (desarrollo)

a) Sobre el cateto vertical $x = 0$, entonces:

$$d\vec{l} = dy\hat{\jmath}$$
$$\vec{F} = I\int d\vec{l} \times \vec{B}$$
$$\vec{F} = 8A\int_0^{1,2m} dy\hat{\jmath} \times (y)0,089\frac{T}{m}\hat{k}$$
$$\vec{F} = 0,513N\hat{\imath}$$

Sobre el lado horizontal $y = 0$, entonces::

$$d\vec{l} = -dx\hat{\imath}$$
$$\vec{F} = I\int d\vec{l} \times \vec{B}$$
$$\vec{F} = 8A\int_0^{1,2m} -dx\hat{\imath} \times (x)0,089\frac{T}{m}\hat{k}$$
$$\vec{F} = 0,513N\hat{\jmath}$$

Para la hipotenusa:

$$y = -x + 1,2$$
$$dy/dx = -1$$
$$d\vec{l} = dx\hat{\imath} + dy\hat{\jmath}$$
$$d\vec{l} = dx\hat{\imath} - dx\hat{\jmath}$$

Entonces:

$$\vec{F} = I\int d\vec{l} \times \vec{B}$$
$$\vec{F} = I\int (dx\hat{\imath} - dx\hat{\jmath}) \times (x + y)0,089\frac{T}{m}\hat{k}$$

Puesto que $x + y = 1{,}2m$

$$\vec{F} = (8A)(1{,}2m)(0,089\frac{T}{m}) \int (dx\hat{\imath} - dx\hat{\jmath}) \times \hat{k}$$
$$\vec{F} = -1{,}025N(\hat{\imath} + \hat{\jmath})$$

b)

$$\vec{\mu} = I\vec{A}$$
$$\vec{\mu} = 8A * \frac{1,2m * 1,2m}{2}\widehat{k}$$
$$\vec{\mu} = 5,76A \cdot m^2\hat{k}$$

c)

$$\vec{\tau} = \vec{\mu} \times \vec{B}$$
$$\vec{\mu} = 5,76A \cdot m^2\hat{k}$$
$$\vec{B} = (x+y)0,089\frac{T}{m}\hat{k}$$
$$\vec{\tau} = 0$$

Puesto que ambos vectores son paralelos.

Problema 2

a)

$$\vec{dl}_1 : R\cos\theta\hat{i} + R\operatorname{sen}\theta\hat{j} + 0\hat{k} \quad \vec{r}_1 = -R\cos\theta\hat{i} - R\operatorname{sen}\theta\hat{j} + 0\hat{k}$$
$$\vec{dl}_1 \times \vec{r}_1 = (-Rd\theta\operatorname{sen}\theta\hat{\imath} + Rd\theta\cos\theta\hat{j}) \times (-R\cos\theta\hat{i} - R\operatorname{sen}\theta\delta) = R^2 d\theta\hat{k}$$

Entonces:

$$\vec{B}_1(\vec{r}) = \frac{\mu_0}{4\pi}\int \frac{I_1}{r_1^3}\vec{dl}_1 \times \vec{r}_1 = \frac{\mu_0 I}{4\pi}\int_0^{2\pi}\frac{1}{R^3}R^2 d\theta\hat{k} = \frac{\mu_0 I}{4\pi R}k\int_0^{2\pi} d\theta = \frac{\mu_0 I}{4\pi R}\hat{k}2\pi$$
$$= \frac{\mu_0 I}{2R}\hat{k}$$

Para el alambre recto:

$$\vec{dl_2} \times \vec{r}_2 = (dx\hat{\imath}) \times (-x\hat{i} + R\hat{j}) = Rdx\hat{i} \times \hat{j} = Rdx\hat{k}$$
$$\vec{B}_2(\vec{r}) = \frac{\mu_0}{4\pi}\int \frac{I_2}{r_2^2}\vec{dl_2} \times \vec{r}_2 = \frac{\mu_0 I}{4\pi}\int_{-L/2}^{+L/2}\frac{Rdx\hat{k}}{\left(\sqrt{x^2+R^2}\right)^3} = \frac{\mu_0 IR}{4\pi}\hat{k}\int_{-L/2}^{+L/2}\frac{dx}{(x^2+R^2)^{3/2}}$$
$$\vec{B}_2(\vec{r}) = \frac{\mu_0 I}{4\pi R}\frac{L}{\sqrt{L^2/4+R^2}}\hat{k}$$

El campo total:

$$\vec{B}_T = \vec{B}_1 + \vec{B}_2 = \frac{\mu_0 I}{2R}\hat{k} + \frac{\mu_0 \cdot I}{4\pi R}\frac{L}{\sqrt{L^2/4+R^2}}\hat{k} = 4,09 \times 10^{-8}T\hat{k}$$

b)

$$\vec{v} = v\frac{1}{\sqrt{3}}(\hat{\imath} + \hat{\jmath} - \hat{k}) = \frac{v}{\sqrt{3}}\hat{\imath} + \frac{v}{\sqrt{3}}\hat{\jmath} - \frac{v}{\sqrt{3}}\hat{k}$$

$$\vec{F_B} = q_0\vec{v} \times \vec{B} = -e\left(v_x\hat{\imath} + v_y\hat{\jmath} + v_z\hat{k}\right) \times B_T\hat{k} = +e\frac{v}{\sqrt{3}}B_T\hat{\jmath} - e\frac{v}{\sqrt{3}}B_T\hat{\imath}$$

$$\vec{F_B} = 1,13 \times 10^{-23}N(-\hat{\imath} + \hat{\jmath})$$